EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Significant Digits

1. Round off the following number within three significant figures $-4.735 \times 10^{-6} kg$

(A) $4.74 \times 10^{-6} kg$ (B) $5.00 \times 10^{-6} kg$ (C) $4.70 \times 10^{-6} kg$ (D) $4.73 \times 10^{-6} kg$

PER055

2. The length, breadth and thickness of a block are given by $l = 12\ cm$, $b = 6\ cm$ and $t = 2.45\ cm$. The volume of the block according to the idea of significant figures should be

(A) $1 \times 10^2 cm^3$ (B) $2 \times 10^2 cm^3$ (C) $1.763 \times 10^2 cm^3$ (D) None of these

PER058

Vernier Callipers and Screw Gauge

3. The vernier of a circular scale is divided in to 30 divisions, which coincides with 29 main scale divisions. If each main scale division is $(1/2)^o$, the least count of the instrument is

(A) 0.1' (B) 1' (C) 10' (D) 30'

PER070

4. The pitch of a screw gauge is 0.5 *mm* and there are 100 divisions on it circular scale. The instrument reads +2 circular scale divisions when nothing is put in-between its jaws. In measuring the diameter of a wire, there are 8 divisions on the main scale and 83rd circular scale division coincides with the reference line. Then the diameter of the wire is

(A) 4.05 *mm* (B) 4.405 *mm* (C) 3.05 *mm* (D) 1.25 *mm*

Rankers Academy JEE

PER072

5. The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 *mm* and there are 50 divisions on the circular scale. The reading on the main scale is 2.5 *mm* and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 2%, the relative percentage error in the density is

(A) 0.9% (B) 2.4% (C) 3.1% (D) 4.2%

PER075

Error Analysis in Experiments

6. An experiment measures quantities x, y, z and then t is calculated from the data as $t = \frac{xy^2}{z^3}$. If percentage errors in x, y and z are respectively 1%, 3%, 2%, then percentage error in t is :

(A) 10 % (B) 4 % (C) 7 % (D) 13 %

PER062

7. Two resistances are measured in ohm and is given as :-

$R_1 = 3\Omega \pm 1\%$

$R_2 = 6\Omega \pm 2\%$

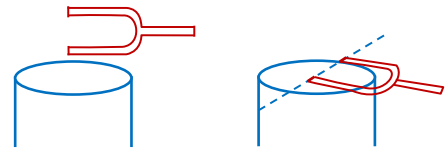
When they are connected in parallel, the percentage error in equivalent resistance is

(A) 3% (B) 4.5% (C) 0.67% (D) 1.33%

PER066

For More Join: @IITJEE_Advanced

8. In a resonance-column experiment, the tuning fork is held above the open end of the glass tube. This can be done in two ways position (1) one prong below the other, and position (2) the two prongs at the same horizontal level. The correct position is –



(A) (1) in all cases
(B) (2) in all cases
(C) (1) for first resonance and (2) for second resonance
(D) (2) for first resonance and (1) for second resonance

PER077

9. A student is performing the experiment of Resonance Column. The diameter of the column tube is 4 *cm*. The distance frequency of the tuning fork is 512 *Hz*. The air temperature is 38°*C* in which the speed of sound is 336 *m/s*. The zero of the meter scale coincides with the top end of the Resonance column. When first resonance occurs, the reading of the water level in the column is

(A) 14.0 (B) 15.2 (C) 16.4 (D) 17.6

PER079

10. Following observation are taken from a travelling microscope to determine the refractive index of a liquid. Reading for the bottom of an empty beaker = 12.324 *cm*. Reading for the bottom of the beaker when partially filled with the liquid = 12.802 *cm*. Reading for the liquid surface = 13.895 *cm*. What is the refractive index of the liquid?

Pankaj's Academy JEE

(A) 1.232 (B) 1.389 (C) 1.437 (D) 1.208

PER081

11. In an optics experiment, with the position of the object fixed, a student varies the position of the convex lens and for each position, the screen is adjusted to get a clear image of the object. A graph between the object distance u and the image distance v, from the lens, is plotted using the same scale for the two axes. A straight line passing through the origin and making an angle of 45° with the x-axis meets the experimental curve at P. The coordinates of P will be:

(A) $\left(\frac{f}{2}, \frac{f}{2}\right)$ (B) (f, f) (C) $(4f, 4f)$ (D) $(2f, 2f)$

PER083

12. A student performs an experiment to determine the Young's modulus of a wire, exactly 2 *m* long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.80 *mm* with an uncertainty of ±0.05 *mm* at a load of exactly 1.0 *kg*. The student also measures the diameter of the wire to be 0.4 *mm* with an uncertainty of ±0.01 *mm*. Take $g = 9.8\ m/s^2$ (exact). The Young's modulus obtained from the reading is

(A) $(2.0 \pm 0.3) \times 10^{11}\ N/m^2$ (B) $(2.0 \pm 0.2) \times 10^{11}\ N/m^2$
(C) $(2.0 \pm 0.1) \times 10^{11}\ N/m^2$ (D) $(2.0 \pm 0.05) \times 10^{11}\ N/m^2$

PER088

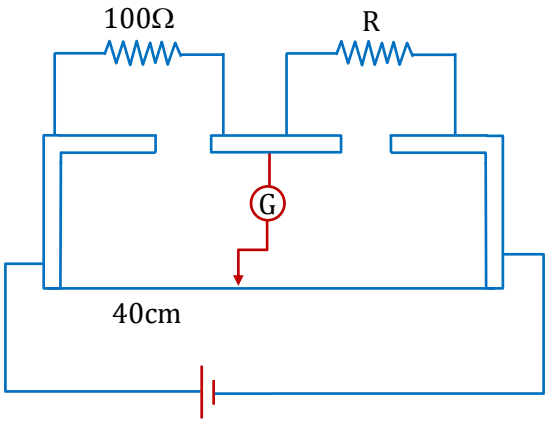
For More Join: @IITJEE_Advanced

13. The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of L is 20.0 *cm* known to 1 *mm* accuracy and time for 100 oscillations of the pendulum is found to be 90*s* using a wrist watch of 1*s* resolution. The accuracy in the determination of g is :

(A) 1% (B) 5% (C) 2% (D) 3%

PER092

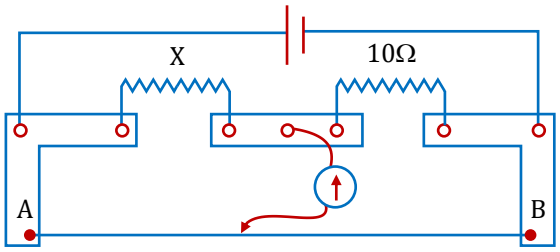
14. In a meter bridge circuit, the known resistance used in resistance box is 100 Ω (without any error, and the unknown resistor is put in right arm). The null point is found to be 40 *cm* from left end. If *mm* scale is used in the meter bridge then resistance of the unknown resistor is:



(A) $\left(150\Omega \pm \frac{3}{8}\right)\Omega$ (B) $\left(150\Omega \pm \frac{1}{8}\right)\Omega$ (C) $\left(150\Omega \pm \frac{5}{8}\right)\Omega$ (D) $\left(150\Omega \pm \frac{7}{16}\right)\Omega$

PER094

15. A meter bridge is set-up as shown, to determine an unknown resistance *'X'* using a standard 10-ohm resistor. The galvanometer shows null point when tapping-key is at 52 *cm* mark. The end-corrections are 1 *cm* and 2 *cm* respectively for the ends *A* and *B*. The determined value of *x* is



(A) 10.2 Ω (B) 10.6 Ω (C) 10.8 Ω (D) 11.1 Ω

PER096

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. In ordinary Vernier callipers, 10th division of the Vernier scale coincides with 9th division of the main scale. In a specially designed Vernier callipers the Vernier scale is so constructed that 10th division on it coincides with 11th division on the main scale. Each division on the main scale equals to 1 *mm*. The callipers have a zero error as shown in the figure-I. When the Vernier calliper is used to measure a length, the concerned portion of its scale is shown in figure-II.

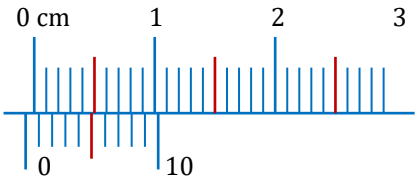


Figure-I

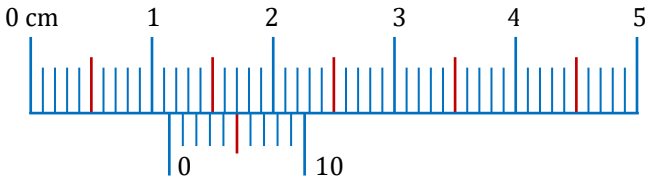


Figure-II

(A) Zero error in the callipers has magnitude 0.7 *mm*.
(B) The length being measured is 1.08 *cm*.
(C) The length being measured is 1.22 *cm*.
(D) Zero error in the callipers has magnitude 0.3 *mm*

PER076

2. We want to design a carbon resistor in shape of a cylinder. Its resistivity is known precisely. The diameter is measured by a screw gauge whose main scale has a least count of 1 *mm* and circular scale has 100 divisions. The length is measured by a vernier callipers whose main scale is 1 *mm* and 9 main scale divisions coincide with 10 vernier scale divisions. We wish that the fourth band in the colour code be painted gold. In which of the following screw gauge readings and vernier callipers readings will our goal be achieved?

(A) Screw gauge: msr 2 divisions, 50th csd coinciding, Vernier scale: msr: 1.2 *cm*, 5th vernier division coinciding.
(B) Screw gauge: msr 1 divisions, 0th csd coinciding, Vernier scale: msr: 0.3 *cm*, 3rd vernier division coinciding.
(C) Screw gauge: msr 0 divisions, 40th csd coinciding, Vernier scale: msr: 5 *cm*, 0th vernier division coinciding.
(D) Screw gauge: msr 1 divisions, 10th csd coinciding, Vernier scale: msr: 8 *cm*, 8th vernier division coinciding.

PER099

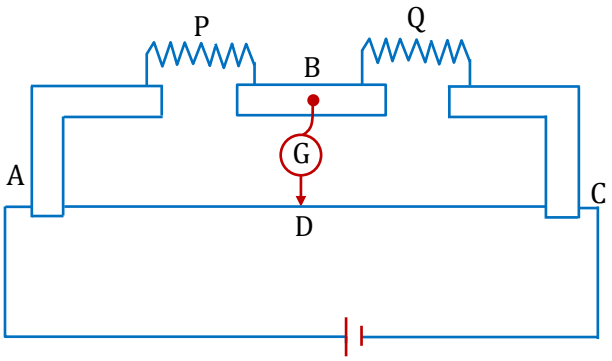
3. In an experiment to measure surface tension of completely wetting liquid, the capillary was dipped in liquid. The level of water in beaker was measured using travelling microscope to be 1.43*cm*. The level of water in capillary was measured to be 6.23*cm*. The readings for inner diameter were 0.00*cm* and 0.32*cm* respectively. Neglect the expansion of capillary and water on heating.

(A) % error in measurement of h is 0.4%
(B) % error in measurement of surface tension is 6.46%
(C) if temperature increases, level of water in capillary will be less than 6.23*cm*.
(D) $s = 0.38 \pm 0.04 N/m$

PER086

For More Join: @IITJEE_Advanced

4. In the meter bridge circuit, the point D is the balance point.



(A) If the jockey is shifted to the left of D, current will flow from the meter bridge wire to the point B.
(B) If jockey is shifted to the right of D, current will flow from point B to the meter bridge wire.
(C) If the jockey is at point D, but now the resistance P is heated, then the current will flow from point B to the meter bridge wire.
(D) If the jockey is at point D, but now the resistance P is heated, then the current will flow from the meter bridge wire to the point B.

PER091

5. Two resistors R_1 and R_2 have measured values as $R_1 = 3.0\ \Omega \pm 0.1\ \Omega$; $R_2 = 6.0\ \Omega \pm 0.2\ \Omega$.
Then
(A) Least count of device used to measure R_2 is $0.2\ \Omega$
(B) When connected in series the error will be 3.34 %
(C) When connected in parallel the error will be $0.1\ \Omega$ (approx.)
(D) When connected in parallel the error will be $0.2\ \Omega$ (approx.)

Bansal's Academy JEE

PER100

6. In the Searle's experiment, after every step of loading, why should we wait for two minutes before taking the readings?
(A) So that the wire can have its desired change in length.
(B) So that the wire can attain room temperature.
(C) So that vertical oscillations can get subsided.
(D) So that the wire has no change in its radius.

PER101

For More Join: @IITJEE_Advanced

EXERCISE (S)

Significant Digits

1. Significant figures are given in the following quantities are p, q, r, s, t, u, v and w respectively.

(A) 343 g (B) 2.20 (C) 1.103 N (D) 0.4142 s

(E) 0.0145 m (F) 1.0080 V (G) 9.1×10^4 km (H) 1.124×10^{-3} V

Value of $\frac{(p+q+r+s+t+u+v+w)}{4}$ is _____

PER056

Vernier Callipers and Screw Gauge

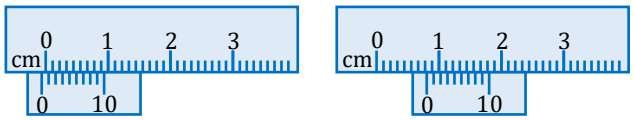
2. The pitch of a screw gauge is 1 mm and there are 50 divisions on its cap. When nothing is put in between the studs, 44th division of the circular scale coincides with the reference line zero of the main scale is not visible. When a glass plate is placed between the studs, the main scale reads three divisions and the circular scale reads 26 divisions. Calculate the thickness of the plate.

PER073

3. Using screw gauge, the observation of the diameter of a wire are 1.324, 1.326, 1.334, 1.336 cm respectively. Find the average diameter, the mean error, the relative error and % error.

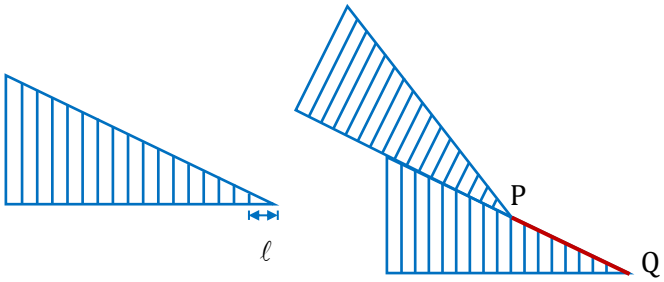
PER104

4. The main scale of a vernier callipers reads in millimeter and its vernier is divided into 10 divisions which coincides with 9 divisions of the main scale. The reading for shown situation is found to be $\frac{23x}{10}$ mm. Find the value of x.



PER105

5. Consider a home made vernier scale as shown in the figure.



In this diagram, we are interested in measuring the length of the line PQ. If both the inclines are identical and their angles are equal to q then what is the least count of the instrument.

PER106

For More Join: @IITJEE_Advanced

Error Analysis in Experiments

6. In a given optical bench, a needle of length 10 *cm* is used to estimate bench error. The object needle, image needle and lens holder have their reading as shown.

$x_0 = 1.1\ cm$ $x_I = 21.0\ cm$ $x_L = 10.9\ cm$

Estimate the bench errors which are present in image needle holder and object needle holder. Also find the focal length of the convex lens when.

$x_0 = 0.6\ cm$ $x_I = 22.5\ cm$ $x_L = 11.4\ cm$

PER085

7. Work done on the system as a function of time t is given as $W_{(t)} = A^3\ln(at^2)$, where $a = 3\ s^{-2}$. The measurement of A has an error of 0.5%. If the error in the measurement of time is 1.25%, the percentage error in the $W_{(t)}$ at $t = 2$ sec is. [Take ln(12) = 2.5]

PER109

8. In surveying, sometimes a value can be found for an inaccessible length ℓ (such as the height of a tall tree) by measuring other three lengths ℓ_1, ℓ_2, ℓ_3 in terms of which $\ell = \frac{\ell_1\ell_2}{\ell_3}$. Suppose we perform such an experiment and obtain the following results (in feet) $\ell_1 = 200 \pm 2$, $\ell_2 = 5.5 \pm 0.1$, $\ell_3 = 10.0 \pm 0.4$. If in this experiment value of ℓ is expressed as $\ell = 110 \pm x$, find value of x (in feet)

PER110

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminum. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and that the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides with the main scale line?

[JEE (Main) 2016]

(1) 0.50 mm (2) 0.75 mm (3) 0.80 mm (4) 0.70 mm

PER010

2. The following observations were taken to determine the surface tension T of water by capillary method:

Diameter of capillary, $D = 1.25 \times 10^{-2} m$

Rise of water, $h = 1.45 \times 10^{-2} m$

Using $g = 9.80\ m/s^2$ and the simplified relation $T = \frac{rhg}{2} \times 10^3\ N/m$, the possible error in surface tension is closest to: **[JEE (Main) 2017]**

(1) 2.4% (2) 10% (3) 0.15% (4) 1.5%

PER012

3. A simple pendulum is being used to determine the value of gravitational acceleration g at a certain place. The length of the pendulum is 25.0 cm and a stop watch with 1s resolution measures the time taken for 40 oscillations to be 50 s. The accuracy in g is: **[JEE (Main) 2020]**

(1) 3.40% (2) 5.40% (3) 4.40% (4) 2.40%

PER014

4. The least count of the main scale of a vernier calipers is 1 mm. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scale. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale. When this vernier is used to measure length of a cylinder the zero of the vernier scale between 3.1 cm and 3.2 cm and 4th VSD coincides with a main scale division. The length of the cylinder is : (VSD is vernier scale division) **[JEE (Main) 2020]**

(1) 3.21 cm (2) 2.99 cm (3) 3.2 cm (4) 3.07 cm

PER015

5. A physical quantity z depends on four observables a, b, c and d, as $z = \frac{a^2 b^{\frac{2}{3}}}{\sqrt{c} d^3}$. The percentage of error in the measurement of a, b, c and d are 2%, 1.5%, 4% and 2.5% respectively. The percentage of error in z is: **[JEE (Main) 2020]**

(1) 12.25% (2) 14.5% (3) 16.5% (4) 13.5%

PER017

For More Join: @IITJEE_Advanced

6. A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 *mm*, 5.55 *mm*, 5.45 *mm*; 5.65 *mm*. The average of these four readings is 5.5375 *mm* and the standard deviation of the data is 0.07395 *mm*. The average diameter of the pencil should therefore be recorded as: **[JEE (Main) 2020]**

(1) (5.5375 ± 0.0739) *mm* (2) (5.538 ± 0.074) *mm*

(3) (5.54 ± 0.07) *mm* (4) (5.5375 ± 0.0740) *mm*

PER018

7. If the length of the pendulum in pendulum clock increases by 0.1%, then the error in time per day is: **[JEE (Main) 2021]**

(1) 86.4 *s* (2) 4.32 *s* (3) 43.2 *s* (4) 8.64 *s*

PER019

8. In the experiment of Ohm's law, a potential difference of 5.0 *V* is applied across the end of a conductor of length 10.0 *cm* and diameter of 5.00 *mm*. The measured current in the conductor is 2.00 *A*. The maximum permissible percentage error in the resistivity of the conductor is :-

[JEE (Main) 2021]

(1) 3.9 (2) 8.4 (3) 7.5 (4) 3.0

PER020

9. A student determined Young's Modulus of elasticity using the formula $Y = \frac{MgL^3}{4bd^3\delta}$. The value of g is taken to be 9.8 m/s^2, without any significant error, his observation are as following.

Physical Quantity	Least Count of the Equipment used for measurement	Observed value
Mass (M)	1 *g*	2 *kg*
Length of bar (L)	1 *mm*	1 *m*
Breadth of bar (b)	0.1 *mm*	4 *cm*
Thickness of bar (d)	0.01 *mm*	0.4 *cm*
Depression (δ)	0.01 *mm*	5 *mm*

Rankers Academy JEE

Then the fractional error in the measurement of *Y* is: **[JEE (Main) 2021]**

(1) 0.0083 (2) 0.0155 (3) 0.155 (4) 0.083

PER021

10. A cylindrical wire of mass $(0.4\pm0.01)g$ has length $(8\pm0.04)cm$ and radius $(6\pm0.03)mm$. The maximum error in its density will be **[JEE (Main) 2023]**

(1) 1% (2) 3.5 % (3) 4% (4) 5%

PER111

11. In an experiment with Vernier callipers of least count 0.1 *mm*, when two jaws are joined together the zero of Vernier scale lies right to the zero of the main scale and 6th division of Vernier scale coincides with the main scale division. While measuring the diameter of a spherical bob, the zero of vernier scale lies in between 3.2 *cm* and 3.3 *cm* marks, and 4th division of vernier scale coincides with the main scale division. The diameter of bob is measured as : **[JEE (Main) 2023]**

(1) 3.18 *cm* (2) 3.25 *cm* (3) 3.26 *cm* (4) 3.22 *cm*

PER112

For More Join: @IITJEE_Advanced

12. In a screw gauge, there are 100 divisions on the circular scale and the main scale moves by 0.5 mm on a complete rotation of the circular scale. The zero of circular scale lies 6 divisions below the line of graduation when two studs are brought in contact with each other. When a wire is placed between the studs, 4 linear scale divisions are clearly visible while 46th division the circular scale coincide with the reference line. The diameter of the wire is ___ $\times 10^{-2}$ mm.

[JEE (Main) 2023]

PER116

13. In an experiment of measuring the refractive index of a glass slab using travelling microscope in physics lab, a student measures real thickness of the glass slab as 5.25 mm and apparent thickness of the glass slab at 5.00 mm. Travelling microscope has 20 divisions in one cm on main scale and 50 divisions on Vernier scale is equal to 49 divisions on main scale. The estimated uncertainty in the measurement of refractive index of the slab is $\frac{x}{10}\times 10^{-3}$, where x is _____.

[JEE (Main) 2023]

PER117

14. In an experiment for estimating the value of focal length of converging mirror, image of an object placed at 40 cm from the pole of the mirror is formed at distance 120 cm from the pole of the mirror. These distances are measured with a modified scale in which there are 20 small divisions in 1 cm. The value of error in measurement of focal length of the mirror is 1/K cm. The value of K is _____ .

[JEE (Main) 2023]

PER118

Rankers Academy JEE

15. In finding out refractive index of glass slab the following observations were made through travelling microscope 50 vernier scale division = 49 MSD; 20 divisions on main scale in each cm

For mark on paper

MSR = 8.45 cm, *VC* = 26

For mark on paper seen through slab

MSR = 7.12 cm, *VC* = 41

For powder particle on the top surface of the glass slab

MSR = 4.05 cm, *VC* = 1

(MSR = Main Scale Reading, *VC* = Vernier Coincidence)

Refractive index of the glass slab is:

[JEE (Main) 2024]

(1) 1.42 (2) 1.52 (3) 1.24 (4) 1.35

PER120

16. There are 100 divisions on the circular scale of a screw gauge of pitch 1 *mm*. With no measuring quantity in between the jaws, the zero of the circular scale lies 5 divisions below the reference line. The diameter of a wire is then measured using this screw gauge. It is found the 4 linear scale divisions are clearly visible while 60 divisions on circular scale coincide with the reference line. The diameter of the wire is :

[JEE (Main) 2024]

(1) 4.65 *mm* (2) 4.55 *mm* (3) 4.60 *mm* (4) 3.35 *mm*

PER121

For More Join: @IITJEE_Advanced

17. Least count of a vernier calliper is $\frac{1}{20N}$ *cm*. The value of one division on the main scale is 1 *mm*. Then the number of divisions of main scale that coincide with *N* divisions of vernier scale is :

[JEE (Main) 2024]

(1) $\left(\frac{2N-1}{20N}\right)$ (2) $\left(\frac{2N-1}{2}\right)$ (3) $(2N-1)$ (4) $\left(\frac{2N-1}{2N}\right)$

PER122

18. Young's modulus is determined by the equation given by $Y = 49000\frac{m}{\ell}\frac{dyne}{cm^2}$ where *M* is the mass and ℓ is the extension of wire used in the experiment. Now error in Young modules(*Y*) is estimated by taking data from *M*-ℓ plot in graph paper. The smallest scale divisions are 5 g and 0.02 *cm* along load axis and extension axis respectively. If the value of *M* and ℓ are 500 g and 2 *cm* respectively then percentage error of *Y* is :

[JEE (Main) 2024]

(1) 0.2 % (2) 0.02 % (3) 2 % (4) 0.5 %

PER123

19. The energy of a system is given as $E(t) = \alpha^3 e^{-\beta t}$, where t is the time and $\beta = 0.3\ s^{-1}$. The errors in the measurement of α and t are 1.2% and 1.6%, respectively. At t = 5 s, maximum percentage error in the energy is :

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[JEE (Main) 2025]

(1) 4% (2) 11.6% (3) 6% (4) 8.4%

PER145

20. A tiny metallic rectangular sheet has length and breadth of 5 mm and 2.5mm, respectively. Using a specially designed screw gauge which has pitch of 0.75 mm and 15 divisions in the circular scale, you are asked to find the area of the sheet. In this measurement, the maximum fractional error will be $\frac{x}{100}$ where *x* is_____

[JEE (Main) 2025]

PER146

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Consider a Vernier callipers in which each 1 *cm* on the main scale is divided into 8 equal divisions and a screw gauge with 100 divisions on its circular scale. In the Vernier callipers, 5 divisions of the Vernier scale coincide with 4 divisions on the main scale and in the screw gauge, one complete rotation of the circular scale moves it by two divisions on the linear scale. Then :

[JEE (Advanced) 2015]

(A) If the pitch of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.01 *mm*.

(B) If the pitch of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.005 *mm*.

(C) If the least count of the linear scale of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.01 *mm*.

(D) If the least count of the linear scale of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.005 *mm*.

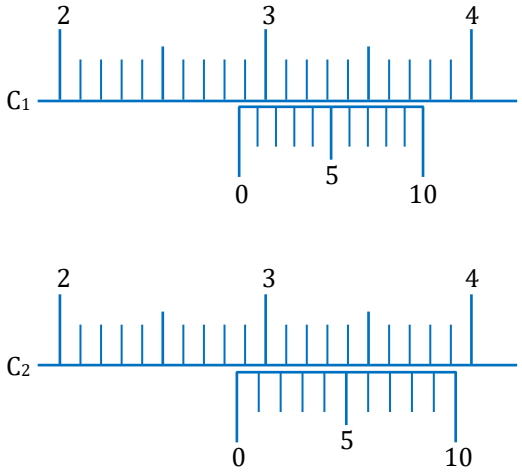
PER043

2. The energy of a system as a function of time t is given as $E(t) = A^2 \exp(-\alpha t)$, where $\alpha = 0.2s^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%, the percentage error in the value of $E(t)$ at $t = 5\ s$ is. **[JEE (Advanced) 2015]**

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PER044

3. There are two vernier calipers both of which have 1 *cm* divided into 10 equal divisions on the main scale. The Vernier scale of one of the calipers (C_1) has 10 equal divisions that correspond to 9 main scale divisions. The Vernier scale of the other caliper (C_2) has 10 equal divisions that correspond to 11 main scale divisions. The readings of the two calipers are shown in the figure. The measured values (in *cm*) by calipers C_1 and C_2 respectively, are **[JEE (Advanced) 2016]**



(A) 2.87 and 2.86 (B) 2.87 and 2.87 (C) 2.87 and 2.83 (D) 2.85 and 2.82

PER045

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4. In an experiment to determine the acceleration due to gravity g, the formula used for the time period of a periodic motion is $T = 2\pi\sqrt{\frac{7(R-r)}{5g}}$. The values of R and r are measured to be (60± 1) mm and (10 ± 1) mm, respectively. In five successive measurements, the time period is found to be 0.52 s, 0.56 s, 0.57s, 0.54 s and 0.59 s. The least count of the watch used for the measurement of time period is 0.01 s. Which of the following statement(s) is(are) true ?

[JEE (Advanced) 2016]

(A) The error in the measurement of r is 10%.
(B) The error in the measurement of T is 3.57%.
(C) The error in the measurement of T is 2%.
(D) The error in the determined value of g is 11%.

PER046

5. A person measures the depth of a well by measuring the time interval between dropping a stone and receiving the sound of impact with the bottom of the well. The error in his measurement of time is δT = 0.01 second and he measures the depth of the well to be L = 20 meters. Take the acceleration due to gravity g = 10 ms^{-2} and the velocity of sound is 300 ms^{-1} . Then the fractional error in the measurement, δL / L, is closest to ; **[JEE (Advanced) 2017]**

(A) 0.2 % (B) 5 % (C) 3 % (D) 1 %

PER047

PARAGRAPH FOR QUESTION NO. 6 and 7

If the measurement errors in all the independent quantities are known, then it is possible to determine the error in any dependent quantity. This is done by the use of series expansion and truncating the expansion at the first power of the error. For example, consider the relation $z = x/y$. If the errors in x, y and z are Δx, Δy and Δz, respectively, then **[JEE (Advanced) 2018]**

$$z \pm \Delta z = \frac{x \pm \Delta x}{y \pm \Delta y} = \frac{x}{y}\left(1 \pm \frac{\Delta x}{x}\right)\left(1 \pm \frac{\Delta y}{y}\right)^{-1}$$

The series expansion for $\left(1 \pm \frac{\Delta y}{y}\right)^{-1}$, to first power in $\Delta y/y$, is $1 \mp (\Delta y/y)$. The relative errors in independent variables are always added. So the error in z will be

$$\Delta z = z\left(\frac{\Delta x}{x} + \frac{\Delta y}{y}\right)$$

The above derivation makes the assumption that $\frac{\Delta x}{x} << 1$, $\frac{\Delta y}{y} << 1$. Therefore, the higher powers of these quantities are neglected.

6. Consider the ratio $r = \frac{(1-a)}{(1+a)}$ to be determined by measuring a dimensionless quantity a. If the error in the measurement of a is $\Delta a (\Delta a/a << 1)$, then what is the error Δr in determining r?

(A) $\frac{\Delta a}{(1+a)^2}$ (B) $\frac{2\Delta a}{(1+a)^2}$ (C) $\frac{2\Delta a}{(1-a^2)}$ (D) $\frac{2a\Delta a}{(1-a^2)}$

PER048

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7. In an experiment the initial number of radioactive nuclei is 3000. It is found that 1000 ± 40 nuclei decayed in the first 1.0 s. For $|x| << 1$, $\ln(1 + x) = x$ up to first power in x. The error $\Delta\lambda$, in the determination of the decay constant λ, in s^{-1}, is :-

(A) 0.04 (B) 0.03 (C) 0.02 (D) 0.01

PER049

8. A steel wire of diameter 0.5 mm and Young's modulus $2 \times 10^{11}\ Nm^{-2}$ carries a load of mass M. The length of the wire with the load is 1.0 m. A vernier scale with 10 divisions is attached to the end of this wire. Next to the steel wire is a reference wire to which a main scale, of least count 1.0 mm, is attached. The 10 divisions of the vernier scale correspond to 9 divisions of the main scale. Initially, the zero of vernier scale coincides with the zero of main scale. If the load on the steel wire is increased by 1.2 kg, the vernier scale division which coincides with a main scale division is..... (Take $g = 10\ ms^{-2}$ and $\pi = 3.2$). **[JEE (Advanced) 2018]**

PER050

9. An optical bench has 1.5 m long scale having four equal divisions in each cm. While measuring the focal length of a convex lens, the lens is kept at 75 cm mark of the scale and the object pin is kept at 45 cm mark. The image of the object pin on the other side of the lens overlaps with image pin that is kept at 135 cm mark. In this experiment, the percentage error in the measurement of the focal length of the lens is. **[JEE (Advanced) 2019]**

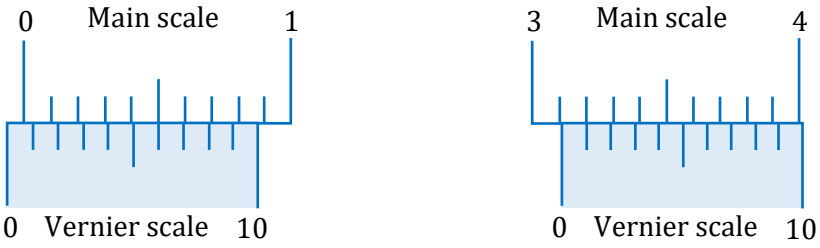
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PER051

10. Two capacitors with capacitance values $C_1 = 2000 \pm 10\ pF$ and $C_2 = 3000 \pm 15\ pF$ are connected in series. The voltage applied across this combination is $V = 5.00 \pm 0.02\ V$. The percentage error in the calculation of the energy stored in this combination of capacitors is _____. **[JEE (Advanced) 2020]**

PER052

11. The smallest division on the main scale of a Vernier calipers is 0.1 cm. Ten divisions of the Vernier scale correspond to nine divisions of the main scale. The figure below on the left shows the reading of this calipers with no gap between its two jaws. The figure on the right shows the reading with a solid sphere held between the jaws. The correct diameter of the sphere is **[JEE (Advanced) 2021]**



(A) 3.07 cm (B) 3.11 cm (C) 3.15 cm (D) 3.17 cm

PER053

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12. Area of the cross-section of a wire is measured using a screw gauge. The pitch of the main scale is 0.5 *mm*. The circular scale has 100 divisions and for one full rotation of the circular scale, the main scale shifts by two divisions. The measured readings are listed below.

Measurement condition	Main scale reading	Circular scale reading
Two arms of gauge touching each other without wire	0 division	4 division
Attempt-1: With wire	4 divisions	20 divisions
Attempt-2: With wire	4 divisions	16 divisions

What are the diameter and cross-sectional area of the wire measured using the screw gauge?

[JEE (Advanced) 2022]

(A) $2.22 \pm 0.02\ mm, \pi(1.23 \pm 0.02)\ mm^2$ (B) $2.22 \pm 0.01\ mm, \pi(1.23 \pm 0.01)\ mm^2$

(C) $2.14 \pm 0.02\ mm, \pi(1.14 \pm 0.02)\ mm^2$ (D) $2.14 \pm 0.01\ mm, \pi(1.14 \pm 0.01)\ mm^2$

PER054

13. In an experiment for determination of the focal length of a thin convex lens, the distance of the object from the lens is $10 \pm 0.1\ cm$ and the distance of its real image from the lens is $20 \pm 0.2\ cm$. The error in the determination of focal length of the lens is n %. The value of n is _____.

[JEE (Advanced) 2023]

PER113

14. The dimensions of a cone are measured using a scale with a least count of 2 mm. The diameter of the base and the height are both measured to be 20.0 cm. The maximum percentage error in the determination of the volume is _____.

[JEE (Advanced) 2024]

PER124

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	B	B	C	D	D	A	B	C
Que.	11	12	13	14	15					
Ans.	D	B	D	C	B					

EXERCISE (O-2)

Que.	1	2	3	4	5	6
Ans.	AC	BC	ACD	ABD	ABC	ABC

EXERCISE (S)

1. 7 **2.** $R_t = 3.64\ mm$

3. $\bar{D} = 1.330 cm . \overline{\Delta D} = 0.005 cm$, Relative error = + 0.004 %, error = 0.4%

4. 3 **5.** $L.C. = \ell\left[\frac{1-\cos\theta}{\cos\theta}\right]$ **6.** 5.5 ± 0.1 cm

7. 2.50 **8.** 8

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	4	3	4	2	3	3	1	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	220	41	32	1	2	2	3	3	3

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9
Ans.	BC	4	C	ABD	D	B	C	3[2.99,3.01]	1.35 to 1.45
Que.	10	11	12	13	14				
Ans.	1.30	C	C	1	3				

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